

A decorative border surrounds the central text, featuring a repeating pattern of stylized skulls, bones, and four-pointed star-like diamonds. The skulls are rendered in a sketchy, hand-drawn style with hollow eye sockets. The bones are simple, elongated shapes. The diamonds are small, four-pointed stars with internal lines. The entire border is set against a solid black background.

SUMMARIZER

Fausto Nunes - 20/03/2023

TABLE OF CONTENTS

01

JDBC

What we use to connect to a database

02

CONNECTION

How we connect to the database

03

STATEMENTS

Provides methods to execute SQL Queries

04

SQL INJECTION

Let's break stuff

05

SOME CODE

A few lines from the JavaBank exercise

A decorative border surrounds the central text, featuring a repeating pattern of stylized skulls, bones, and four-pointed stars (diamonds) in a light blue color against a dark blue background.

01

JAVA DATABASE CONNECTIVITY

JAVA DATABASE CONNECTIVITY

Although the Master Coders are still torturing us
by not giving us tools that do this almost
automatically, let's review this before they give
us the GOOD STUFF!

FIRST IMPRESSIONS

"WHY AM I DOING THIS MANUALLY?"

Well...:

- We might need to optimize databases in the future
- If something goes wrong we now know how it is being done under the hood



JDBC



IT'S AN API

It's a way for our Java programs to have access to database management systems



WORKS WITH DIFFERENT DBs

JDBC has drivers for a variety of different databases

The background is a dark blue gradient. It is decorated with numerous white line-art icons of human skulls and long bones (femurs) scattered across the surface. Interspersed among these are small, four-pointed star or diamond shapes. The central text is white and stands out against the dark background.

02

CONNECTION

How we actually connect to a database

CONNECTION



DRIVERMANAGER

Connects an application to a data source, which is specified by a database URL

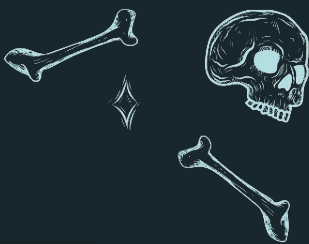
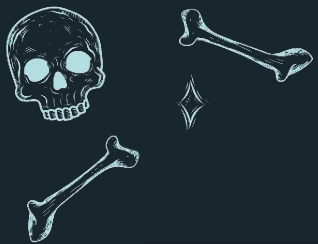


DATABASE URL

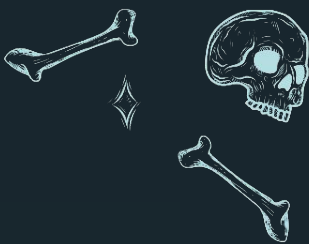
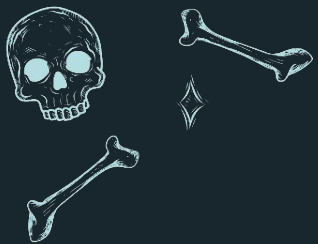
The long and weird URL with lots of optional stuff. You know the one...



JDBC:MYSQL://[HOST][,FAILOVERHOST...]:PORT/
[DATABASE]?PROPERTYNAME[=PROPERTYVALUE]



```
private Connection connection = null;  
  
connection = DriverManager.getConnection(dbUrl, user, pass);
```



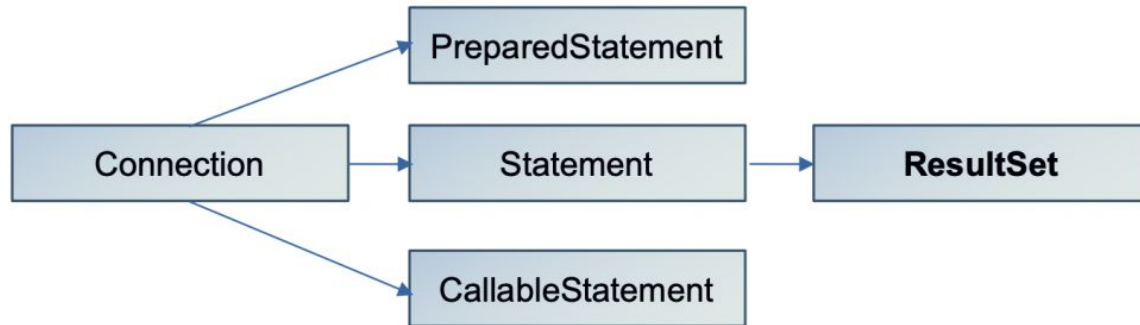
```
public void close() {  
    try {  
        if (connection != null) {  
            connection.close();  
        }  
    } catch (SQLException ex) {  
        System.out.println("Failure to close database connections: " + ex.getMessage());  
    }  
}
```



A decorative border surrounds the central text, featuring a repeating pattern of stylized skulls, bones, and four-pointed stars (diamonds) in a light blue color against a dark blue background.

03

STATEMENTS



STATEMENT RELATED STUFF

STATEMENT

It executes normal SQL statements with no IN and OUT parameters.



PREPARED STATEMENT

It executes parameterized SQL statement that supports IN parameter.

STATEMENT RELATED STUFF



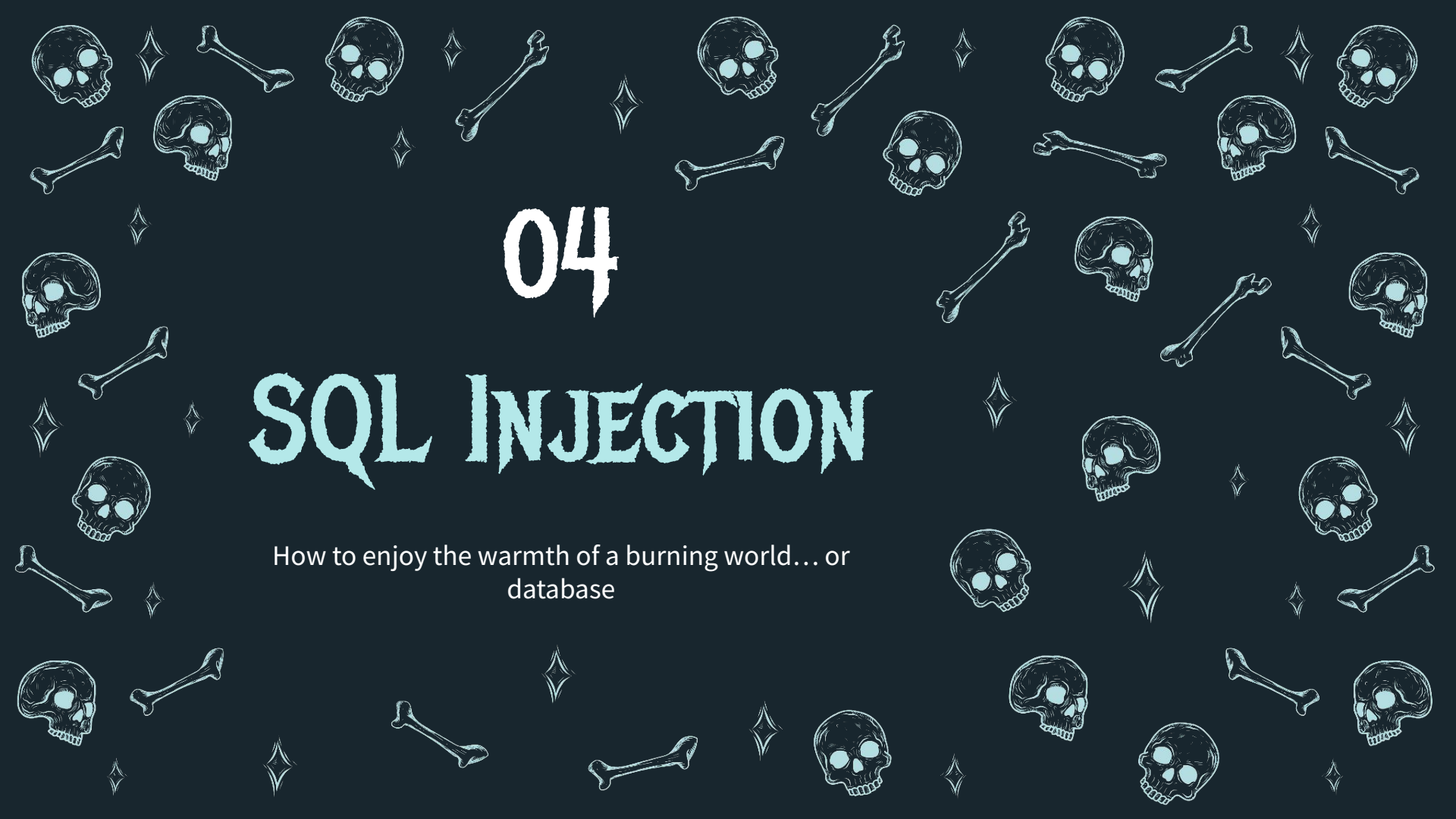
CALLABLE STATEMENT

Used to execute stored procedures that may contain both input and output parameters.



RESULTSET

ResultSet maintains cursor/pointer which points to a single row of the query results. Using navigational and getter methods provided by ResultSet, we can iterate and access database records one by one.

The background is a dark blue gradient. It is decorated with numerous white line-art icons of human skulls, long bones, and four-pointed stars (diamonds) scattered across the entire surface. The skulls are shown from various angles, some facing forward and others in profile. The bones are long and thin, resembling femurs. The diamonds are small and simple line drawings.

04

SQL INJECTION

How to enjoy the warmth of a burning world... or
database

FUNCTIONING OF AN SQL INJECTION

1. Hacker identifies vulnerable, SQL-driven website & injects malicious SQL query via input data.

01



WEBSITE INPUT FIELDS

2. Malicious SQL query is validated & command is executed by database.

02



DATABASE

3. Hacker is granted access to view and alter records or potentially act as database administrator

03



HACKER

A decorative border surrounds the central text, featuring stylized white skulls, bones, and four-pointed stars (diamonds) on a black background.

05 SOME CODE

JavaBank we meet again...



```
@Override
public Customer get(Integer id) {
    try {

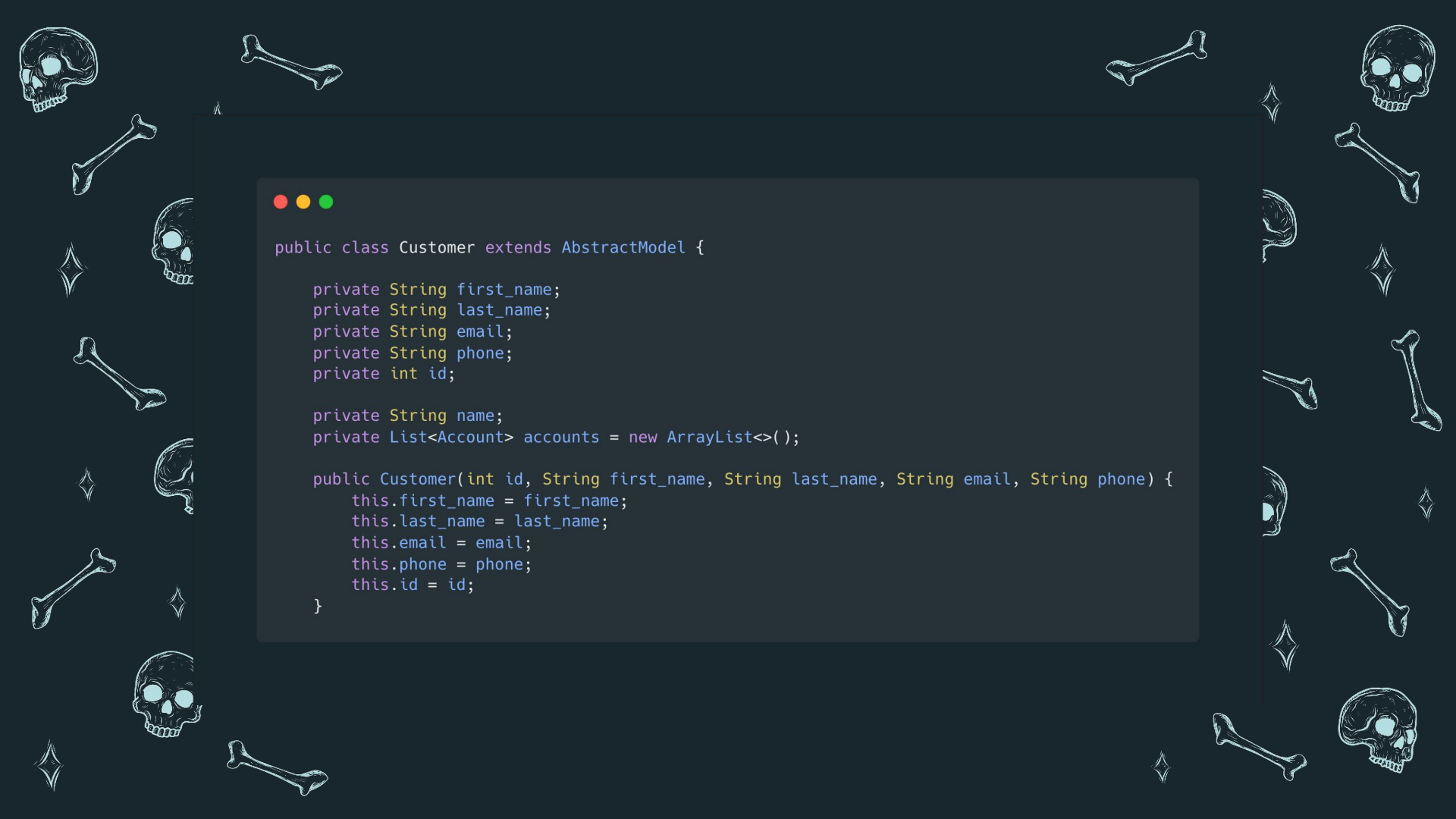
        Statement statement = connection.createStatement();
        String query = "SELECT * FROM customer WHERE id = id";
        ResultSet resultSet = statement.executeQuery(query);

        if (resultSet.next()) {
            int result = resultSet.getInt(id);
            String firstName = resultSet.getString("first_name");
            String lastName = resultSet.getString("last_name");
            String email = resultSet.getString("email");
            String phone = resultSet.getString("phone");

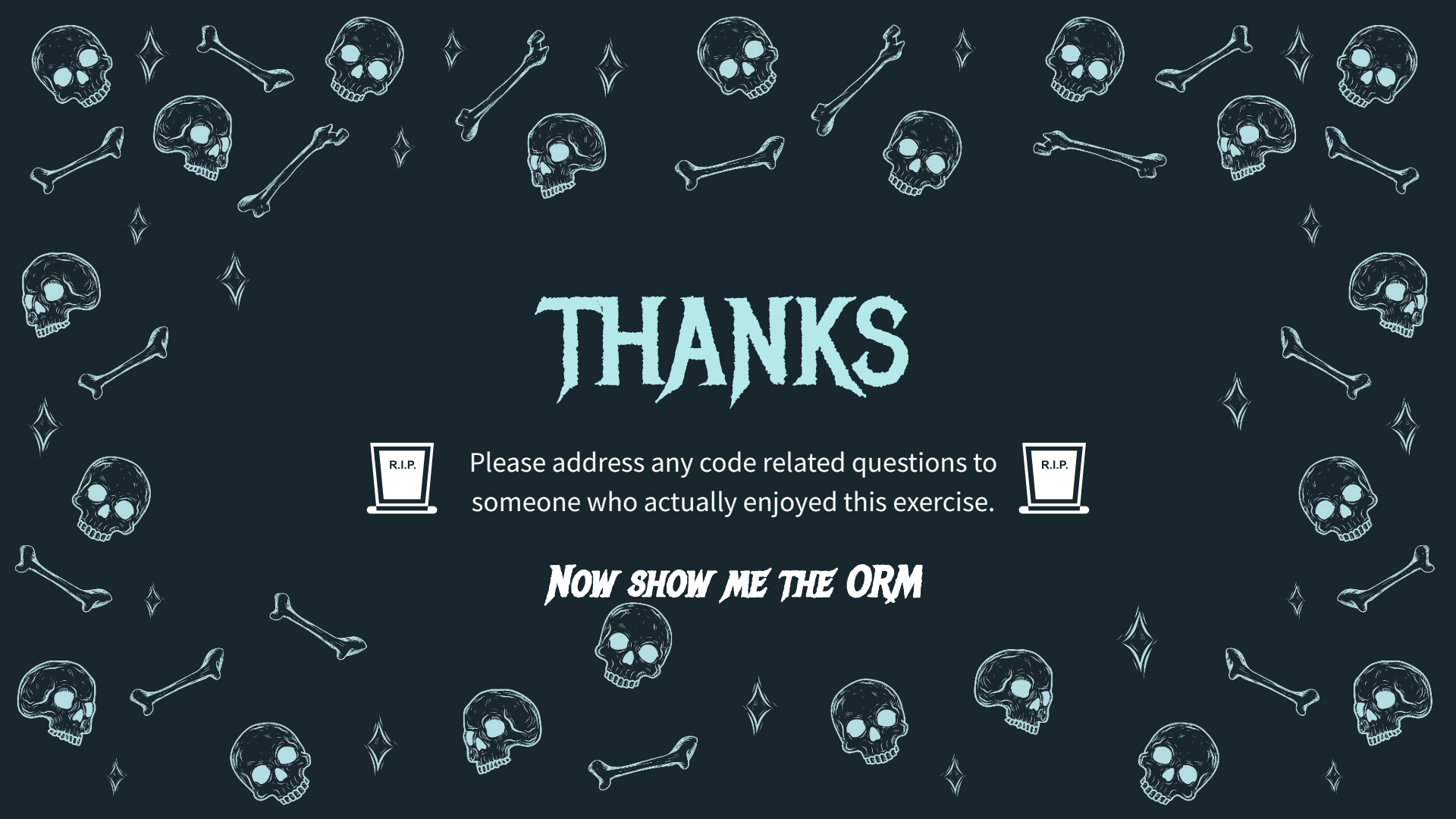
            customer = new Customer(result, firstName, lastName, email, phone);
        }

    } catch (SQLException e) {
        throw new RuntimeException(e);
    }

    return customer;
}
```



```
public class Customer extends AbstractModel {  
  
    private String first_name;  
    private String last_name;  
    private String email;  
    private String phone;  
    private int id;  
  
    private String name;  
    private List<Account> accounts = new ArrayList<>();  
  
    public Customer(int id, String first_name, String last_name, String email, String phone) {  
        this.first_name = first_name;  
        this.last_name = last_name;  
        this.email = email;  
        this.phone = phone;  
        this.id = id;  
    }  
}
```

The entire slide is framed by a decorative border. It features a repeating pattern of stylized white skulls, long bones, and four-pointed star-like diamonds on a dark blue background. The elements are scattered around the perimeter, creating a macabre, Halloween-themed frame.

THANKS



Please address any code related questions to
someone who actually enjoyed this exercise.



NOW SHOW ME THE ORM